MSG No. 224.113 $Pn\bar{3}m'$ [Type III, cubic]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'3m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]`

Table 2: Wyckoff site: `4b`, site symmetry: `.-3m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,6,13,17,18,32,34,36,44,46,48]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,10,11,14,22,23,28,29,33,40,41,45]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,7,12,15,19,24,25,30,35,37,42,47]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,9,16,20,21,26,27,31,38,39,43]`

Table 3: Wyckoff site: `4c`, site symmetry: `.-3m`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,5,6,13,17,18,32,34,36,44,46,48]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,10,11,14,22,23,28,29,33,40,41,45]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,7,12,15,19,24,25,30,35,37,42,47]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[4,8,9,16,20,21,26,27,31,38,39,43]`

Table 4: Wyckoff site: `6d`, site symmetry: `-4'2.m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,2,3,4,37,38,45,46]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,8,10,12,41,42,43,44]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,7,9,11,39,40,47,48]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[13,14,15,16,25,26,33,34]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[17,20,22,24,29,30,31,32]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[18,19,21,23,27,28,35,36]`

Table 5: Wyckoff site: `8e`, site symmetry: `.3m'`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6,44,46,48]`
2	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[2,10,11,40,41,45]`
3	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	`[3,7,12,37,42,47]`
4	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	`[4,8,9,38,39,43]`
5	$[-x, -x, -x]$	`[13,17,18,32,34,36]`
6	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[14,22,23,28,29,33]`
7	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	`[15,19,24,25,30,35]`

8	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	`[16,20,21,26,27,31]`

Table 6: Wyckoff site: `12f`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2,33,34]`
2	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,25,26]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,12,30,32]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,9,27,36]`
5	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[7,11,28,35]`
6	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,10,29,31]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,14,45,46]`
8	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[15,16,37,38]`
9	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[17,24,42,44]`
10	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[18,21,39,48]`
11	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[19,23,40,47]`
12	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[20,22,41,43]`

Table 7: Wyckoff site: `12g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,45,46]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	`[3,4,37,38]`
3	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,12,42,44]`
4	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[6,9,39,48]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[7,11,40,47]`
6	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[8,10,41,43]`
7	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[13,14,33,34]`
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[15,16,25,26]`
9	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[17,24,30,32]`
10	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[18,21,27,36]`
11	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	`[19,23,28,35]`
12	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[20,22,29,31]`

Table 8: Wyckoff site: `24h`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4]`
3	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[5,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[6,9]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[7,11]`
6	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[8,10]`
7	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[15,16]`

continued ...

Table 8

No.	position	mapping
9	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	[17,24]
10	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[18,21]
11	$\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[19,23]
12	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[20,22]
13	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[25,26]
14	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[27,36]
15	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[28,35]
16	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[29,31]
17	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[30,32]
18	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[33,34]
19	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[37,38]
20	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	[39,48]
21	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[40,47]
22	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[41,43]
23	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	[42,44]
24	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[45,46]

Table 9: Wyckoff site: 24i, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ y+\frac{1}{2}\right]$	[1,33]
2	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -y\right]$	[2,34]
3	$[0,\ y,\ -y]$	[3,26]
4	$\left[0,\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[4,25]
5	$\left[y+\frac{1}{2},\ \frac{1}{2},\ y\right]$	[5,30]
6	$\left[y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[6,27]
7	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ 0\right]$	[7,28]
8	$[-y,\ 0,\ y]$	[8,29]
9	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{2}\right]$	[9,36]
10	$\left[y+\frac{1}{2},\ 0,\ \frac{1}{2}-y\right]$	[10,31]
11	$[y,\ -y,\ 0]$	[11,35]
12	$\left[-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[12,32]
13	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-y\right]$	[13,45]
14	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y\right]$	[14,46]
15	$[0,\ -y,\ y]$	[15,38]
16	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[16,37]
17	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -y\right]$	[17,42]
18	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[18,39]
19	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[19,40]
20	$[y,\ 0,\ -y]$	[20,41]
21	$\left[y+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	[21,48]
22	$\left[\frac{1}{2}-y,\ 0,\ y+\frac{1}{2}\right]$	[22,43]
23	$[-y,\ y,\ 0]$	[23,47]
24	$\left[y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[24,44]

Table 10: Wyckoff site: `24j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{2}, y, -y\right]$	`[1,34]`
2	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	`[2,33]`
3	$\left[0, y, y+\frac{1}{2}\right]$	`[3,25]`
4	$\left[0, \frac{1}{2}-y, -y\right]$	`[4,26]`
5	$\left[-y, \frac{1}{2}, y\right]$	`[5,32]`
6	$\left[y, -y, \frac{1}{2}\right]$	`[6,36]`
7	$\left[\frac{1}{2}-y, -y, 0\right]$	`[7,35]`
8	$\left[y+\frac{1}{2}, 0, y\right]$	`[8,31]`
9	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	`[9,27]`
10	$\left[-y, 0, \frac{1}{2}-y\right]$	`[10,29]`
11	$\left[y, y+\frac{1}{2}, 0\right]$	`[11,28]`
12	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	`[12,30]`
13	$\left[\frac{1}{2}, -y, y\right]$	`[13,46]`
14	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	`[14,45]`
15	$\left[0, -y, \frac{1}{2}-y\right]$	`[15,37]`
16	$\left[0, y+\frac{1}{2}, y\right]$	`[16,38]`
17	$\left[y, \frac{1}{2}, -y\right]$	`[17,44]`
18	$\left[-y, y, \frac{1}{2}\right]$	`[18,48]`
19	$\left[y+\frac{1}{2}, y, 0\right]$	`[19,47]`
20	$\left[\frac{1}{2}-y, 0, -y\right]$	`[20,43]`
21	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	`[21,39]`
22	$\left[y, 0, y+\frac{1}{2}\right]$	`[22,41]`
23	$\left[-y, \frac{1}{2}-y, 0\right]$	`[23,40]`
24	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	`[24,42]`

Table 11: Wyckoff site: `24k`, site symmetry: `..m'`

No.	position	mapping
1	$[x, x, z]$	`[1,44]`
2	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[2,41]`
3	$\left[\frac{1}{2}-x, x, \frac{1}{2}-z\right]$	`[3,42]`
4	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[4,43]`
5	$[z, x, x]$	`[5,48]`
6	$[x, z, x]$	`[6,46]`
7	$\left[\frac{1}{2}-x, z, \frac{1}{2}-x\right]$	`[7,37]`
8	$\left[\frac{1}{2}-z, \frac{1}{2}-x, x\right]$	`[8,39]`
9	$\left[\frac{1}{2}-x, \frac{1}{2}-z, x\right]$	`[9,38]`
10	$\left[z, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[10,40]`
11	$\left[x, \frac{1}{2}-z, \frac{1}{2}-x\right]$	`[11,45]`
12	$\left[\frac{1}{2}-z, x, \frac{1}{2}-x\right]$	`[12,47]`
13	$[-x, -x, -z]$	`[13,32]`
14	$\left[-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$	`[14,29]`
15	$\left[x+\frac{1}{2}, -x, z+\frac{1}{2}\right]$	`[15,30]`
16	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[16,31]`
17	$[-z, -x, -x]$	`[17,36]`

continued ...

Table 11

No.	position	mapping
18	$[-x,\ -z,\ -x]$	[18,34]
19	$\left[x+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[19,25]
20	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[20,27]
21	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[21,26]
22	$\left[-z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[22,28]
23	$\left[-x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23,33]
24	$\left[z+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[24,35]

Table 12: Wyckoff site: 48l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[y,\ z,\ x]$	[6]
7	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[7]
8	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[8]
9	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[9]
10	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[10]
11	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[11]
12	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[16]
17	$[-z,\ -x,\ -y]$	[17]
18	$[-y,\ -z,\ -x]$	[18]
19	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[19]
20	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[20]
21	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[21]
22	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23]
24	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\ -z,\ y+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -y\right]$	[26]
27	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ -x\right]$	[27]
28	$\left[-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[28]
29	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[30]
31	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[31]
32	$[-y,\ -x,\ -z]$	[32]
33	$\left[-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[33]
34	$[-x,\ -z,\ -y]$	[34]

continued ...

Table 12

No.	position	mapping
35	$\left[z+\frac{1}{2},\,-y,\,x+\frac{1}{2}\right]$	[35]
36	$[-z,\,-y,\,-x]$	[36]
37	$\left[\frac{1}{2}-x,\,z,\,\frac{1}{2}-y\right]$	[37]
38	$\left[\frac{1}{2}-x,\,\frac{1}{2}-z,\,y\right]$	[38]
39	$\left[\frac{1}{2}-z,\,\frac{1}{2}-y,\,x\right]$	[39]
40	$\left[z,\,\frac{1}{2}-y,\,\frac{1}{2}-x\right]$	[40]
41	$\left[y,\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$	[41]
42	$\left[\frac{1}{2}-y,\,x,\,\frac{1}{2}-z\right]$	[42]
43	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,z\right]$	[43]
44	$[y,\,x,\,z]$	[44]
45	$\left[x,\,\frac{1}{2}-z,\,\frac{1}{2}-y\right]$	[45]
46	$[x,\,z,\,y]$	[46]
47	$\left[\frac{1}{2}-z,\,y,\,\frac{1}{2}-x\right]$	[47]
48	$[z,\,y,\,x]$	[48]